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ME469: Review of Partial Differential Equations

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Equation Review

Real World

- The intended application

Conceptual Model

- Physical laws, hypothesis and models (e.g. a set of PDEs) and initial and boundary conditions

Computer Model

- A set of computational algorithms that allow to build a numerical, or approximated solution to the conceptual model

Partial Differential Equation: Transient Evolution of a Scalar with Advection, Diffusion, and Source

General Transport Equation using Einstein notation, i.e., repeated index, for scalar ϕ

- Density, ρ [kg/m^3]
- Velocity, u_j [m/s]
- Mass diffusivity, D [m^2/s^2]
- Dynamic viscosity, μ [kg/(m-s)]
- Schmidt number, Sc

$$\frac{\partial \rho\phi}{\partial t} + \frac{\partial \rho u_j \phi}{\partial x_j} - \frac{\partial}{\partial x_j}\left(\rho D \frac{\partial \phi}{\partial x_j}\right) = S^{\phi} \qquad \rho D = \frac{\mu}{Sc}$$

$$\underbrace{\frac{\partial \rho\phi}{\partial t}}_{\text{Time term, or “accumulation”}} + \underbrace{\frac{\partial \rho u_j \phi}{\partial x_j}}_{\text{advection}} - \underbrace{\frac{\partial}{\partial x_j}\left(\frac{\mu}{Sc} \frac{\partial \phi}{\partial x_j}\right)}_{\text{diffusion}} = \underbrace{S^{\phi}}_{\text{source/sink term}}$$

Repeated Index Example

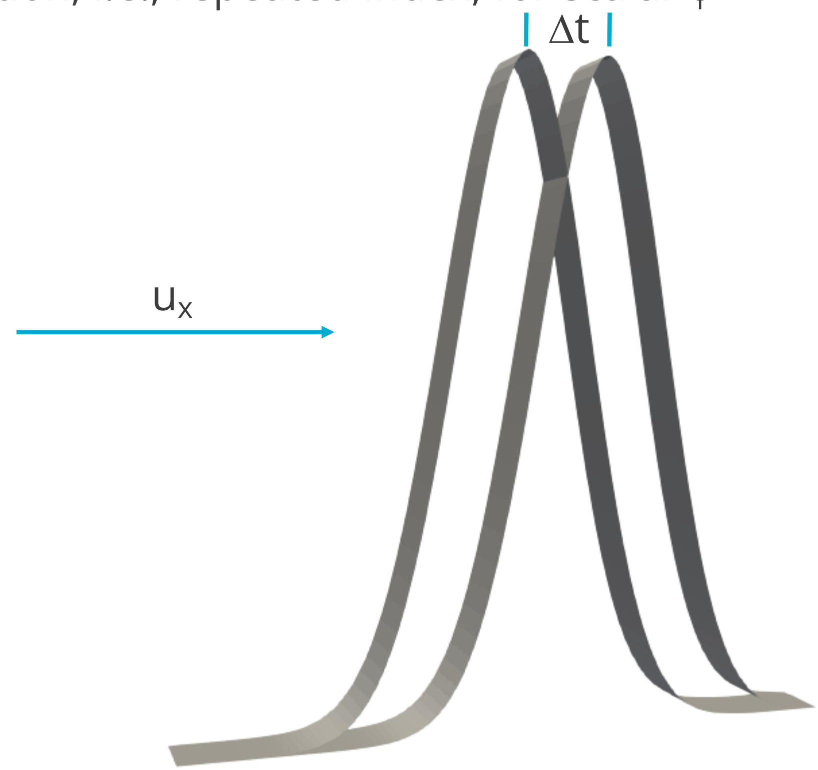
$$\frac{\partial \rho u_j \phi}{\partial x_j} = \frac{\partial \rho u_x \phi}{\partial x} + \frac{\partial \rho u_y \phi}{\partial y} + \frac{\partial \rho u_z \phi}{\partial z}$$

Partial Differential Equation: Transient Evolution of a Scalar with Advection

General Transport Equation using Einstein notation, i.e., repeated index, for scalar ϕ

- Density, ρ [kg/m^3]
- Velocity, u_j [m/s]
- Mass diffusivity, D [m^2/s^2]
- Dynamic viscosity, μ [kg/(m-s)]
- Schmidt number, Sc

$$\frac{\partial \rho \phi}{\partial t} + \frac{\partial \rho u_j \phi}{\partial x_j} - \cancelto{0.0}{\frac{\partial}{\partial x_j}\left(\frac{\mu}{Sc}\frac{\partial \phi}{\partial x_j}\right)} = \cancelto{0.0}{S^{\phi}}$$

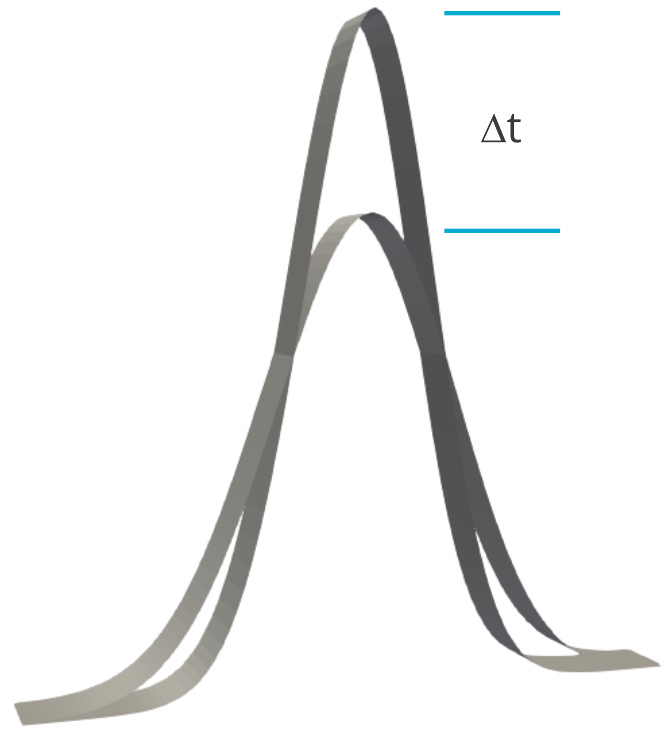


Partial Differential Equation: Transient Evolution of a Scalar with Diffusion

General Transport Equation using Einstein notation, i.e., repeated index, for scalar ϕ

- Density, ρ [kg/m^3]
- Velocity, u_j [m/s]
- Mass diffusivity, D [m^2/s^2]
- Dynamic viscosity, μ [kg/(m-s)]
- Schmidt number, Sc

$$\frac{\partial \rho \phi}{\partial t} + \cancelto{0.0}{\frac{\partial \rho u_j \phi}{\partial x_j}} - \frac{\partial}{\partial x_j}\left(\frac{\mu}{Sc}\frac{\partial \phi}{\partial x_j}\right) = \cancelto{0.0}{S^{\phi}}$$



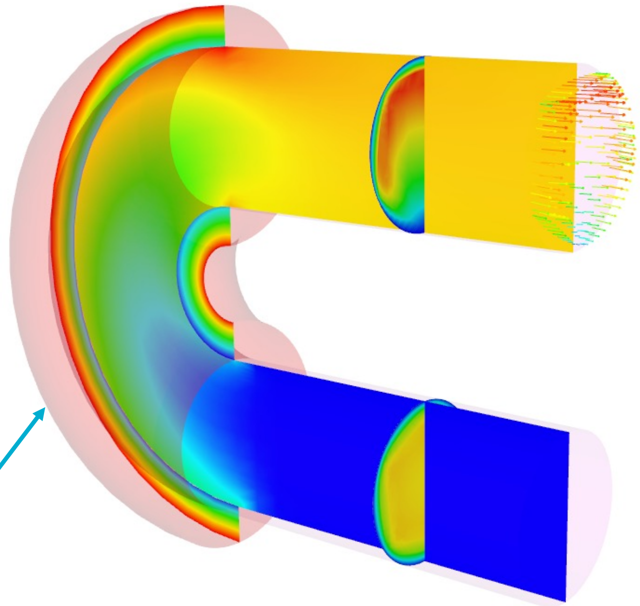
Partial Differential Equation: Transient Heat Conduction

Thermal Heat Conduction

- Density, ρ [kg/m^3]
- Specific Heat, C_p [J/(kg-K)]
- Thermal conductivity, λ [W/(m-K)], 1W = 1 J/s
- Heat flux vector, q_j

$$\rho C_p \frac{\partial T}{\partial t} + \frac{\partial q_j}{\partial x_j} = S$$

$$q_j = -\lambda \frac{\partial T}{\partial x_j}$$



Thermal heat conduction (outer shell heated)

Variable Density, non-Isothermal Fluid Flow System

- Consider the variable density (non-isothermal) equations of motion (momentum and continuity) with energy transport:

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho u_j}{\partial x_j} = 0,$$

$$\frac{\partial \rho u_i}{\partial t} + \frac{\partial \rho u_j u_i}{\partial x_j} + \frac{\partial P}{\partial x_i} = \frac{\partial \tau_{ij}}{\partial x_j} + \rho g_i,$$

$$\frac{\partial \rho E}{\partial t} + \frac{\partial \rho u_j H}{\partial x_j} = -\frac{\partial q_j}{\partial x_j} + \frac{\partial u_i \tau_{ij}}{\partial x_j} + \rho u_i g_i$$

(2+nDim)

$\rho = \frac{PM}{RT}$ Equation of State (EOS) provides the P and ρ relationship

Constitutive Relationships

$$E = H - P/\rho,$$

$$H = h + \frac{1}{2} u_k u_k,$$

$$\tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} \mu \frac{\partial u_k}{\partial x_k} \delta_{ij},$$

$$q_i = -k \frac{\partial T}{\partial x_i},$$

$$h = \int_{T_o}^{T} C_p dT$$

Conserved vs Non-conserved Form

For the so-called non-conserved form, chain-rule the advection term, while enforcing continuity:

$$\frac{\partial \rho \phi}{\partial t} + \frac{\partial \rho u_j \phi}{\partial x_j} - \frac{\partial}{\partial x_j}\left(\frac{\mu}{Sc}\frac{\partial \phi}{\partial x_j}\right) = S^{\phi}$$

$$\rho \frac{\partial \phi}{\partial t} + \rho u_j \frac{\partial \phi}{\partial x_j} + \phi \underbrace{\left(\frac{\partial \rho}{\partial t} + \frac{\partial \rho u_j}{\partial x_j}\right)}_{0.0} - \frac{\partial}{\partial x_j}\left(\frac{\mu}{Sc}\frac{\partial \phi}{\partial x_j}\right) = S^{\phi}$$

$$\longrightarrow \quad \rho \frac{\partial \phi}{\partial t} + \rho u_j \frac{\partial \phi}{\partial x_j} - \frac{\partial}{\partial x_j}\left(\frac{\mu}{Sc}\frac{\partial \phi}{\partial x_j}\right) = S^{\phi}$$

Incompressible Navier Stokes

For a constant density and viscosity (in absence of a source term), we revert to the classic form:

$$\cancelto{0.0}{\frac{\partial \rho}{\partial t}} + \frac{\partial \rho u_j}{\partial x_j} = 0,$$

$$\frac{\partial \rho u_i}{\partial t} + \frac{\partial \rho u_j u_i}{\partial x_j} + \frac{\partial P}{\partial x_i} = \frac{\partial \tau_{ij}}{\partial x_j}$$

$$\tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} \mu \cancelto{0.0}{\frac{\partial u_k}{\partial x_k}} \delta_{ij},$$

$$\frac{\partial u_j}{\partial x_j} = 0,$$

$$\rho \frac{\partial u_i}{\partial t} + \rho \frac{\partial u_j u_i}{\partial x_j} - \mu \frac{\partial^2 u_i}{\partial {x_j}^2} = -\frac{\partial P}{\partial x_i}$$

(trick)

$$\frac{\partial}{\partial x_j} \mu \frac{\partial u_j}{\partial x_i} = \mu \frac{\partial}{\partial x_i} \cancelto{0.0}{\frac{\partial u_j}{\partial x_j}}$$